

ABDUL WAQAR

Artificial Intelligence Engineer

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SUMMARY

I am a BS Computer Science (Specialization in Artificial Intelligence) graduate from NED University. I have hands-on experience in Machine Learning, Deep Learning, Computer Vision, and NLP. I am skilled in Python, SQL, and React Native, and I have completed internships at Sapphire Consulting Services, NIKUD, PIA, and PCSIR. My projects include a GAN-based Synthetic Data Generation system, a from-scratch GPT-2 implementation, Hate Speech Detection, a Smart Parking system using YOLO, and a Library Recommendation System. Certified Generative AI Developer and participant at Indus AI Week 2026.

SKILLS

- Machine Learning:** Machine Learning, Deep Learning, Computer Vision, Natural Language Processing (NLP), Image Processing, YOLO, OpenCV, PyTorch, TensorFlow/Keras, TorchVision, GANs, CNN, Transformers (BERT/GPT), Hugging Face, LLMs, RAG, LangChain, Data Mining
- Languages & Frameworks:** Python, C, C++, SQL, React Native, Node.js, Streamlit, HTML, CSS
- Tools & Cloud:** Git, GitHub, Firebase, Google Colab, Scikit-learn, Ollama, CUDA/GPUs, MLflow/W&B, Databases (Pandas, NumPy, Database Management)
- Core Competencies:** Problem Solving, Team Collaboration

CERTIFICATIONS AND ACHIEVEMENTS

- Generative AI Application Developer Training - NCEAC-HEC (iCodeGuru, PakAngels, Aspire Pakistan, UETIANS Lahore) | 2026
- Python Programming - edX | 2025
- Indus AI Week - NYLP | Feb 2026 - Participated in a 3-day national conference covering AI advancements, Women in AI, AI for All, and Youth in AI

EXPERIENCE

Sapphire Consulting Services - AI Engineering Intern

Jul 2026 - Aug 2026

Karachi, Pakistan

- Built a DCGAN to generate synthetic images, addressing class imbalance for the underrepresented class in a classification dataset.
- Trained a CNN classifier on the GAN-augmented dataset, improving accuracy on the minority class.
- Developed a Streamlit app to generate new images and display the Discriminator's real/fake confidence score in real time.

NIKUD - React Native and Full Stack Intern

Jul 2025 - Aug 2025

Karachi, Pakistan

- Developed a cross-platform mobile application using React Native and Firebase for Android.
- Built an Admin Dashboard using Node.js for real-time data visualization and user management.
- Integrated RESTful APIs to connect the mobile app with the backend dashboard.
- Implemented Firebase Authentication and Realtime Database for enhanced security and data retrieval.

Pakistan International Airlines (PIA) - React Native Intern

Aug 2024 - Sep 2024

Karachi, Pakistan

- Developed cross-platform mobile applications using React Native and Firebase.
- Implemented real-time database integration for improved app responsiveness.
- Collaborated with senior developers to debug and optimize code, reducing app crashes.

Pakistan Council of Scientific and Industrial Research (PCSIR) - Data Analyst Intern

Sep 2023 - Oct 2023

Karachi, Pakistan

- Performed data cleaning and preprocessing on large datasets, handling missing values and inconsistencies.
- Improved data accuracy through systematic validation and outlier detection.
- Assisted in data mining tasks to increase reporting efficiency.

PROJECTS

Synthetic Data Generation for Face Classification | Python, PyTorch, DCGAN, CNN, Scikit-learn, Streamlit

- Built a DCGAN to generate realistic synthetic images, addressing class imbalance in a classification dataset.
- Trained a CNN classifier on the augmented dataset to improve performance on the underrepresented class.
- Built a Streamlit app to generate new images and display the Discriminator's real/fake confidence score.

GPT-2 Implementation from Scratch | Python, NumPy

- Implemented a GPT-2 style transformer from scratch in NumPy, including tokenization, embeddings, masked multi-head self-attention, and backpropagation.
- Trained a character-level language model using AdamW, and implemented autoregressive text generation with temperature control.

Hate Speech and Cyberbullying Detection | Python, NLP, Scikit-learn, Streamlit

- Built an AI classifier for Hate Speech, Cyberbullying, or Neutral text using TF-IDF and Logistic Regression.
- Combined machine learning with rule-based keyword filtering for low-confidence cases.
- Developed an interactive Streamlit app with real-time analysis, confidence scores, and confusion matrix visualization.

Smart Park - Object Detection for Parking Space Monitoring | Python, YOLO, Streamlit

- Built a real-time parking space monitoring system using YOLO to classify filled and empty spaces in images and video.
- Implemented confidence thresholding and color-coded bounding boxes for intuitive visualization.
- Deployed via a Streamlit interface supporting media upload and real-time detection.

Library Recommendation System | Python, NLP, TF-IDF, Streamlit

- Built a content-based book recommendation engine using TF-IDF vectorization and cosine similarity on book descriptions.
- Integrated fuzzy matching to handle user input typos and spelling errors.
- Deployed as a Streamlit app with Plotly visualizations for top recommendations by rating.

General Health Query Chatbot | Python, Groq API, GPT-OSS 20B, Streamlit

- Built a multi-turn health chatbot with a Streamlit interface using GPT-OSS 20B via the Groq API.
- Applied prompt engineering and safety rules to prevent diagnosis, medication advice, and unsafe self-harm responses.

EDUCATION

NED University of Engineering and Technology

Bachelor of Computer Science - Specialization in Artificial Intelligence
CGPA: 3.17

2022 - 2026

Karachi, Pakistan

Govt. Degree Science/Commerce College, Gulshan-e-Iqbal

Intermediate - 81.09%

2020 - 2022

Karachi, Pakistan

Everest Grammar High School

Matriculation - 82.5%

2018 - 2020

Karachi, Pakistan

LANGUAGES

- English – Intermediate
- Urdu – Native
- Sindhi – Native